

IE 306 — Systems Simulation

Term Project: Istanbul Ferry Terminal Operations (Group Project — Teams of 3)

Spring 2026 | Due Date: Last day of finals

Overview

Istanbul's Bosphorus ferry network carries hundreds of thousands of commuters daily between the Asian and European sides of the city. In this project your team will build a discrete-event simulation of a **four-terminal ferry network** using SimPy. The network connects two Asian-side terminals (Kadıköy and Üsküdar) to two European-side terminals (Eminönü and Beşiktaş) via four regular ferry lines and one optional shuttle line.

Your model must faithfully represent **time-varying passenger demand**, **schedule-based ferry operations** with berth contention, **batch boarding** with dwell-time limits, **route switching** for left-behind passengers, and **weather disruptions**. You will design and run experiments comparing operational interventions (increased frequency, a European-side shuttle line) under both normal and storm conditions.

Unlike the assignments, this project requires you to make and defend your own modeling decisions. The system description below defines *what* must be modeled; *how* you implement each aspect in SimPy is up to you. Your report should justify your design choices.

1 System Description

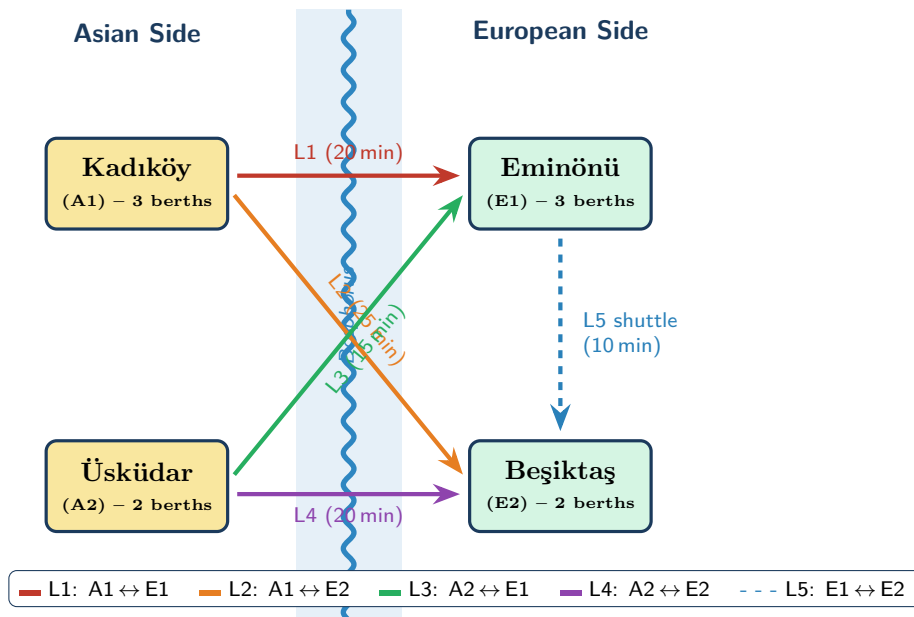


Figure 1: Bosphorus ferry network. Four terminals connected by four regular lines (L1–L4) and one optional shuttle (L5, dashed). Only cross-Bosphorus OD pairs are served by the ferry system.

1.1 Terminal Network

The system consists of four terminals on two sides of the Bosphorus (see Figure 1). Each terminal has a finite number of **berths** (docking positions for ferries), a **waiting area** with finite passenger capacity, and **turnstiles** controlling entry.

Table 1: Terminal parameters.

Terminal	Code	Berths	Waiting cap.	Turnstiles	Dwell time
Kadıköy	A1	3	1 500 pax	8	6 min
Üsküdar	A2	2	600 pax	4	5 min
Eminönü	E1	3	1 200 pax	6	6 min
Beşiktaş	E2	2	800 pax	4	5 min

Berths represent docking slots. A ferry arriving at a terminal with all berths occupied must **wait offshore** until a berth becomes available.

Waiting area. Passengers who have passed through the turnstiles wait in the terminal’s shared waiting area. If the number of waiting passengers equals the waiting capacity, the next passenger exiting the turnstile **balks** (leaves the system immediately and is counted as lost). In other words, a passenger passes through the turnstile first and *then* discovers the waiting area is full.

Dwell time. When a ferry docks, it remains at the berth for a fixed *dwell time* (see Table 1). During this time passengers board. Boarding stops when the ferry reaches capacity *or* the dwell time expires, whichever comes first. Passengers who could not board (*left behind*) remain in the waiting area for the next sailing.

1.2 Ferry Lines

Five ferry lines operate on the network. Lines L1–L4 are always active; line L5 (the European-side shuttle) is active only in shuttle scenarios.

Table 2: Ferry line parameters.

Line	Route	Type	Capacity	Peak hdwy	Off-peak hdwy	Travel time
L1	A1 ↔ E1	Large vapur	400 pax	15 min	30 min	20 min
L2	A1 ↔ E2	Small motor	200 pax	20 min	40 min	25 min
L3	A2 ↔ E1	Small motor	200 pax	20 min	40 min	15 min
L4	A2 ↔ E2	Small motor	200 pax	30 min	60 min	20 min
L5	E1 ↔ E2	Small motor	200 pax	15 min	15 min	10 min

All lines operate **bidirectionally**: scheduled departures occur from both endpoints independently. Peak headways apply during **07:00–09:00** and **17:00–19:00**; off-peak headways apply during all other hours. The first departure on each line (in each direction) occurs at **06:00**.

Headway transitions. When a period boundary is crossed, the next departure uses the new period’s headway measured from the *previous departure time* (not from the boundary itself). For example, if the last off-peak departure was at 06:50 and peak begins at 07:00 with a 15-min headway, the next departure is at $06:50 + 15 = 07:05$.

1.3 Passenger Arrivals

Passengers arrive at each terminal as a **Poisson process** with a **time-varying rate**. Only **cross-Bosphorus** origin–destination (OD) pairs are served; passengers travelling between same-

side terminals (e.g. Kadıköy to Üsküdar) do not use the ferry system and are not modeled.

Time-of-day periods. The operating day is divided into four periods based on commuter patterns:

Table 3: Time-of-day periods.

Period	Hours
AM peak	07:00–09:00
Midday	09:00–17:00
PM peak	17:00–19:00
Low	06:00–07:00, 19:00–22:00

1.3.1 Terminals with Historical Data (A1 and E1)

For **Kadıköy (A1)** and **Eminönü (E1)**, six months (182 days) of historical arrival records are provided in `arrivals_kadikoy.csv` and `arrivals_eminonu.csv`. Each row represents one passenger arrival:

Table 4: Historical data format.

Column	Type	Description
<code>date</code>	YYYY-MM-DD	Calendar date
<code>time</code>	HH:MM:SS.fff	Arrival time of day
<code>destination</code>	string	Destination terminal name

Example rows from `arrivals_kadikoy.csv`:

```
date,time,destination
2025-01-01,06:00:06.867,Eminonu
2025-01-01,06:00:16.551,Besiktas
```

Destination values are ASCII terminal names: `Eminonu`, `Besiktas`, `Kadikoy`, `Uskudar`.

Your task. Using the period definitions in Table 3, perform a complete **input analysis** for each terminal–period combination:

1. Compute inter-arrival times within each period.
2. Fit an **exponential distribution** to the inter-arrival times (i.e. estimate λ for each period). Include histograms and Q–Q plots.
3. Run a **goodness-of-fit test** (e.g. Kolmogorov–Smirnov) and report the test statistic and p -value. Discuss whether the exponential assumption is reasonable.
4. Estimate the **destination split** (fraction of passengers to each destination) from the data.
5. Use your fitted rates and destination splits in your simulation model—these are the values that go into your `env.timeout(rng.exponential(1/λ))` calls.

Connecting fitted rates to the rate formula. The effective-rate formula in §1.3.3 uses a single *base rate* λ_o^{base} scaled by direction multipliers. For A1 and E1 you may either (a) use your per-period fitted rates directly as the effective rate for each period (the simplest approach), or (b) estimate a single base rate from the AM-peak period (where the dominant-direction multiplier is 1.0) and apply the formula with multipliers. Both approaches are acceptable; just be consistent and document your choice.

Present your full input analysis in your report (see §6).

1.3.2 Terminals with Given Parameters (A2 and E2)

For **Üsküdar (A2)** and **Beşiktaş (E2)**, arrival rates and destination splits are given directly:

Table 5: Arrival parameters for A2 and E2 (given).

Origin	Base rate (pax/hr)	Destination 1 (%)	Destination 2 (%)
Üsküdar (A2)	1 000	55% → Eminönü	45% → Beşiktaş
Beşiktaş (E2)	500	60% → Kadıköy	40% → Üsküdar

These are **base rates** corresponding to the dominant-direction peak (the period where $m_{\text{dir}} = 1.0$: AM peak for Asian-side terminals, PM peak for European-side terminals). Apply the direction multipliers from Table 6 to obtain the effective rate at time t .

1.3.3 Effective Arrival Rate

The ferry system exhibits a strong *tidal* commuter pattern: morning traffic flows predominantly from Asia to Europe, and evening traffic reverses. The effective arrival rate at time t for a passenger from origin o to destination d is:

$$\lambda_{o \rightarrow d}(t) = \lambda_o^{\text{base}} \times p_{o \rightarrow d} \times m_{\text{dir}}(\text{period}(t), \text{direction}(o, d))$$

where λ_o^{base} is the base arrival rate (fitted from data for A1/E1, or given for A2/E2), $p_{o \rightarrow d}$ is the destination split, and m_{dir} is the direction multiplier from Table 6. Direction is *Asia*→*Europe* or *Europe*→*Asia*.

Table 6: Direction multipliers by time period.

Period	Asia → Europe	Europe → Asia
AM peak	1.0	0.3
Midday	0.4	0.4
PM peak	0.3	1.0
Low	0.15	0.15

Example. Suppose your fitted base rate for Kadıköy is $\hat{\lambda}_{A1} = 2000$ pax/hr and the fitted destination split to Eminönü is $\hat{p} = 0.65$. Then during AM peak the effective A1→E1 rate is $2000 \times 0.65 \times 1.0 = 1300$ pax/hr.

1.4 Terminal Service

Turnstile entry. Each terminal has a number of turnstiles (Table 1). A passenger must pass through a turnstile to enter the waiting area. Turnstile service time is $\text{Exp}(\text{mean} = 3 \text{ s})$ per passenger. Multiple turnstiles serve passengers simultaneously.

Boarding. Passengers board a ferry one at a time, with boarding time $\text{Exp}(\text{mean} = 1 \text{ s})$ per passenger. Boarding from a terminal’s waiting area is **first-come-first-served** among passengers whose *next leg* is the ferry’s line (this includes transfer passengers waiting for their second leg). Boarding continues until the ferry reaches its capacity or the dwell time expires, whichever occurs first.

1.5 Ferry Operations

Each ferry line follows a **deterministic schedule** defined by its headway (Table 2). At each scheduled departure:

1. The ferry requests a **berth** at the origin terminal. If all berths are occupied, the ferry waits offshore.
2. Once docked, the ferry remains for the terminal’s **dwelt time**. During this time, waiting passengers board (see §1.4).
3. After dwell time expires, the ferry departs. Passengers who could not board remain in the waiting area.
4. The ferry **travels** to the destination terminal for the line’s travel time.
5. The ferry requests a berth at the destination, disembarks all passengers (instantaneous), releases the berth, and the cycle repeats.

Simplifying assumption (no vessel tracking). Ferries are *not* modeled as individual persistent objects—each scheduled departure is an independent process with its own capacity. In reality a single vessel shuttles back and forth: if its outbound sailing is cancelled, it never reaches the far terminal, so the return sailing cannot operate either. Our model deliberately omits this coupling. Cancellation draws (Section 1.7) are independent per sailing and per direction, and each scheduled departure “spawns” its own ferry regardless of what happened to earlier sailings. This is a conscious scope decision—the simulation focuses on *passenger flow and terminal performance*, not fleet management or vessel rotation.

1.6 Route Choice and Left-Behind Passengers

Each passenger is assigned a **destination** upon arrival (Table 5 and your fitted splits for A1/E1). The passenger’s **primary route** is the direct ferry line connecting their origin to their destination:

Table 7: Primary route assignment.

Origin	Destination	Direct line
A1 (Kadıköy)	E1 (Eminönü)	L1
A1 (Kadıköy)	E2 (Beşiktaş)	L2
A2 (Üsküdar)	E1 (Eminönü)	L3
A2 (Üsküdar)	E2 (Beşiktaş)	L4
E1 (Eminönü)	A1 (Kadıköy)	L1
E1 (Eminönü)	A2 (Üsküdar)	L3
E2 (Beşiktaş)	A1 (Kadıköy)	L2
E2 (Beşiktaş)	A2 (Üsküdar)	L4

Route switching. Passengers in the waiting area do *not* predict whether they will fit on an arriving ferry; they wait until the ferry actually departs. When a ferry departs full and a passenger is *left behind*, the following rule applies **only in shuttle scenarios** (when L5 is active):

- If an **indirect route via the shuttle** exists for the passenger’s OD pair, the passenger switches to the indirect route with probability $q = 0.6$. With probability $1 - q = 0.4$ the passenger stays and waits for the next direct sailing.
- A switching passenger is **removed** from the current terminal’s waiting area and begins a **two-leg journey**: first the cross-Bosphorus leg to Eminönü (or Beşiktaş), then the shuttle (L5) to the final destination.
- Transfer passengers re-enter the transfer terminal’s waiting area through the turnstiles (subject to turnstile service and waiting-area capacity).

Indirect routes enabled by the shuttle:

Table 8: Indirect routes via shuttle (L5).

OD pair	Leg 1 (cross)	Leg 2 (shuttle)
A1 → E2	L1: A1 → E1	L5: E1 → E2
A2 → E2	L3: A2 → E1	L5: E1 → E2
E2 → A1	L5: E2 → E1	L1: E1 → A1
E2 → A2	L5: E2 → E1	L3: E1 → A2

Only the four OD pairs listed in Table 8 have indirect alternatives. All other left-behind passengers (e.g. A1→E1, A2→E1) have no indirect route and simply wait for the next direct sailing, even when the shuttle is active.

In scenarios **without** the shuttle, left-behind passengers simply wait for the next direct ferry.

1.7 Weather Disruption

Weather conditions are modeled as a **scenario parameter**: the weather state is fixed for an entire replication. Two states exist:

- **Calm**: No disruptions. All scheduled sailings operate normally.
- **Lodos** (strong southerly wind): Each individual scheduled sailing has an independent probability $p_{\text{cancel}} = 0.20$ of being **cancelled**. A cancelled sailing simply does not occur—the ferry skips that departure and passengers wait for the next scheduled service. Cancellation decisions are made **per sailing, per direction**: if the 08:00 L2 departure from A1 is cancelled, the 08:00 L2 departure from E2 is *not* automatically cancelled—it has its own independent cancellation draw.

2 Scenarios

Run the following five mandated scenarios:

Table 9: Experimental scenarios.

#	Description	Frequency	Shuttle (L5)	Weather	Purpose
S1	Baseline	Standard	No	Calm	Reference
S2	High frequency	+50%	No	Calm	Capacity investment
S3	Shuttle network	Standard	Yes	Calm	Network design
S4	Storm stress test	Standard	No	Lodos	Disruption impact
S5	Full intervention	+50%	Yes	Lodos	Combined mitigation

“+50% **frequency**” means each headway is multiplied by $2/3$ and rounded to the nearest minute:

Table 10: Headways under +50% frequency (S2 and S5).

Line	Baseline peak	+50% peak	Baseline off-pk	+50% off-pk
L1	15 min	10 min	30 min	20 min
L2	20 min	13 min	40 min	27 min
L3	20 min	13 min	40 min	27 min
L4	30 min	20 min	60 min	40 min
L5 (if active)	15 min	10 min	15 min	10 min

In addition to the five mandated scenarios, your team must design and run **at least one additional scenario (S6 or more)** of your own choosing. Describe its purpose and motivation in your report.

3 Simulation Requirements

3.1 Random Number Streams

Use `numpy.random.SeedSequence` and `default_rng()` to create **separate RNG streams** for each source of randomness (arrivals per origin, turnstile service, boarding, destination assignment, route switching, weather cancellation, etc.). Document your stream assignments clearly.

Common Random Numbers (CRN). Use the **same base seed per replication** across all scenarios so that the same passenger arrival sequences are generated. This ensures fair paired comparisons.

3.2 Simulation Window and Warm-up

- Simulated day: **06:00–22:00** (57 600 s per replication).
- Discard the first **1 hour** (3 600 s, i.e. 06:00–07:00) as warm-up. Do not include observations from this period in any KPI.
- Passengers still in the system when the simulation ends (in turnstiles, waiting areas, or on ferries) are **excluded** from all KPIs. Only fully completed journeys contribute to journey times, wait times, and throughput.
- Number of replications: **20**.
- Report point estimates and **95% confidence intervals** (using the t -distribution with $n - 1$ degrees of freedom) for all KPIs.

4 Key Performance Indicators

For each scenario and replication, compute the following KPIs. The **exact KPI names** in the rightmost column are the strings you must use in your output files (see §5).

Table 11: Required KPIs. All times in seconds; rates and factors as fractions in $[0, 1]$.

Definition	Details	KPI name
Avg. journey time, all passengers	Turnstile entry $\rightarrow$ arrival at destination	avg_journey_time
Avg. journey time $\rightarrow$ Eminönü	Only pax with destination E1	avg_jt_eminonu
Avg. journey time $\rightarrow$ Beşiktaş	Only pax with destination E2	avg_jt_besiktas
Avg. journey time $\rightarrow$ Kadıköy	Only pax with destination A1	avg_jt_kadikoy
Avg. journey time $\rightarrow$ Üsküdar	Only pax with destination A2	avg_jt_uskudar
Passenger loss rate	Fraction of arrivals who balk (terminal full)	loss_rate
Left-behind rate	Pax who miss ≥ 1 ferry / total arrivals (incl. balked)	left_behind_rate
Throughput	Total pax delivered per hour (post-warmup)	throughput
Load factor, L1	Mean (boarded / capacity) per L1 departure that actually sailed	load_factor_L1
Load factor, L2	Same for L2	load_factor_L2
Load factor, L3	Same for L3	load_factor_L3
Load factor, L4	Same for L4	load_factor_L4
Avg. wait at Kadıköy	Enter waiting area $\rightarrow$ board ferry, at A1 (only pax who boarded)	avg_wait_kadikoy
Avg. wait at Üsküdar	Same, A2	avg_wait_uskudar
Avg. wait at Eminönü	Same, E1	avg_wait_eminonu
Avg. wait at Beşiktaş	Same, E2	avg_wait_besiktas
Berth utilization, Kadıköy	Time-avg fraction of berths occupied at A1	berth_util_kadikoy
Berth utilization, Eminönü	Same, E1	berth_util_eminonu
Missed-connection rate	Fraction of transfer pax waiting > 1 L5 headway at transfer terminal; report 0 in scenarios without shuttle	missed_conn_rate
Total passengers served	Count of pax completing journey (post-warmup)	total_pax_served

5 Output Format

Your notebook must produce two CSV files. **The auto-grader reads these files directly—** use the exact column names and KPI strings below.

5.1 results.csv — Per-replication data

One row per (scenario, replication, KPI) combination:

scenario	replication	kpi	value
S1	1	avg_journey_time	1842.3
S1	1	loss_rate	0.032
...	...	...	...
S5	20	total_pax_served	31204

Types: `scenario` (string), `replication` (integer 1–20), `kpi` (string, exact names from Ta-

ble 11), value (float).

5.2 summary.csv — Aggregated statistics

One row per (scenario, KPI), computed across the 20 replications:

scenario	kpi	mean	ci_lower	ci_upper	std
S1	avg_journey_time	1842.3	1790.1	1894.5	112.4
...	...	...	...	...	...

CI computation: use the t -distribution with $n - 1 = 19$ degrees of freedom.

6 Deliverables

Submit a **single folder** named with your team identifier (e.g. `team_01`) containing the three items below. Do **not** include the input data files (`arrivals_*.csv`), virtual environments, or `__pycache__` directories.

1. **Jupyter Notebook** (`.ipynb`)—one file only, containing all simulation code, clearly structured and documented. Must be runnable end-to-end (restart kernel → Run All) and complete within 15 minutes. The notebook will be re-executed by the auto-grader; if it fails to run, you lose execution credit (A1, 10 pts).
2. **Output files:** `results.csv` and `summary.csv` in the format specified above. Include these pre-generated files in your submission as a safety net—if your notebook fails during auto-grading, the pre-generated CSVs will still be graded for A2–A6. If both the notebook fails *and* no CSVs are present, the auto-graded portion scores zero.
3. **Report** (PDF, max 15 pages excluding code appendix):
 - a. **Input analysis** for Kadıköy (A1) and Eminönü (E1): inter-arrival time histograms, Q–Q plots, GoF test results, fitted rates per period, and destination split estimates (see §1.3.1).
 - b. A **conceptual model diagram** showing the passenger lifecycle (arrive → turnstile → wait → board → travel → [transfer?] → arrive) and ferry operations cycle.
 - c. A description of your **model logic and assumptions** beyond what is specified here, including justification of SimPy construct choices.
 - d. **Results tables and/or figures** for all scenarios, with 95% confidence intervals visible.
 - e. A **comparative discussion**: How does increased frequency affect performance? What does the shuttle achieve? How does Iodos degrade the system? Which KPI is most affected?
 - f. A **sensitivity analysis**: Systematically vary at least two factors (e.g. dwell time, waiting capacity, cancellation probability) and discuss which factor has the largest impact.
 - g. A **verification and validation** discussion: What sanity checks did you perform? How did you convince yourselves the model is correct? Include at least three checks with numerical evidence.
 - h. A description and discussion of your **student-designed scenario(s)** (S6+).

7 Grading Rubric

Grading has two components. **Part A** is auto-graded by a script that reads your `results.csv` and `summary.csv`. **Part B** is graded manually from your report.

Table 12: Grading rubric (100 points total).

	Component	Points
Auto (75)	A1. Execution & format compliance	10
	A2. Completeness (scenarios, reps, KPIs)	7
	A3. KPI value accuracy	30
	A4. Statistical methodology (CIs, CRN, warm-up)	12
	A5. Comparative direction checks	10
	A6. Consistency checks	6
Manual	B1. Input analysis (A1 & E1 arrivals)	10
	B2. Report quality (diagram, analysis, S6+)	8
	B3. Sensitivity analysis & V&V	7
Total		100

A3 detail: Your reported KPI means are compared against reference intervals derived from the instructor solution. Values within the reference band receive full credit; values within a wider tolerance band receive partial credit; values outside receive zero.

A4 detail: The auto-grader recomputes confidence intervals from your raw `results.csv` and checks consistency with your `summary.csv`. It also verifies CRN implementation by checking that `total_pax_served` is identical (within 2%) across S1 and S2 for each replication.

A5 detail: The auto-grader checks that your model produces qualitatively correct comparative results (e.g. S2 journey times < S1, S4 loss rate $\geq$ S1).

B1 detail: Your input analysis section must include, for each terminal–period combination (A1 and E1, four periods each): *(i)* inter-arrival time histograms with fitted exponential overlay (4 pts), *(ii)* Q–Q plots for the exponential fit (2 pts), *(iii)* GoF test results with interpretation (2 pts), *(iv)* destination split estimates with justification (2 pts).

Parameter Summary

For easy reference, all numerical parameters are collected below:

Table 13: Complete parameter set.

Parameter	Symbol	Value
<i>Terminals</i>		
Kadıköy berths / wait cap / turnstiles / dwell		3 / 1 500 / 8 / 6 min
Üsküdar berths / wait cap / turnstiles / dwell		2 / 600 / 4 / 5 min
Eminönü berths / wait cap / turnstiles / dwell		3 / 1 200 / 6 / 6 min
Beşiktaş berths / wait cap / turnstiles / dwell		2 / 800 / 4 / 5 min
<i>Ferry lines</i>		
L1 capacity / peak hdwy / off-pk hdwy / travel		400 / 15 min / 30 min / 20 min
L2 capacity / peak hdwy / off-pk hdwy / travel		200 / 20 min / 40 min / 25 min
L3 capacity / peak hdwy / off-pk hdwy / travel		200 / 20 min / 40 min / 15 min
L4 capacity / peak hdwy / off-pk hdwy / travel		200 / 30 min / 60 min / 20 min
L5 (shuttle) cap / hdwy / travel		200 / 15 min / 10 min
<i>Service</i>		
Turnstile service time		Exp(mean = 3 s)
Boarding time per passenger		Exp(mean = 1 s)
<i>Demand</i>		
Kadıköy base rate	λ_{A1}	Fit from data
Eminönü base rate	λ_{E1}	Fit from data
Üsküdar base rate	λ_{A2}	1 000 pax/hr
Beşiktaş base rate	λ_{E2}	500 pax/hr
A1/E1 destination splits		Estimate from data
Route-switch probability	q	0.6
<i>Weather</i>		
Lodos cancellation probability	p_{cancel}	0.20
<i>Simulation</i>		
Simulated day		06:00–22:00 (57 600 s)
Warm-up		3 600 s
Replications		20
Confidence level		95% (t_{19})